

Manraj Dhaliwal

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EDUCATION

University of Michigan - Dearborn

April 2026

Bachelor of Computer Science | GPA: 3.5

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Edge-Computing, Computer Networks, Software Security, Artificial Intelligence, Database Management, Web Systems

EXPERIENCE

Software Engineer Intern

Oct 2025 – Present

Bosch

Farmington Hills, MI

- Built a full-stack web app (React, FastAPI, PostgreSQL, AWS) that converts engine measurement files into interactive dashboards with filterable sensor charts and downloadable reports, reducing processing time by over 90%.
- Developed an automated invoice distribution system using Python (openpyxl, Jinja2) and Microsoft Graph API that took in billing data from Excel and replaced a manual invoicing process for 15+ department managers, dispatching 100+ personalized client invoices monthly, reducing turnaround time by 70% and saving 20+ hours of administrative work per month.
- Built a C++ signal processing pipeline that intakes automotive sensor data, detects signal faults and threshold violations, and generates summary reports for calibration engineers, processing 1M+ data points per run.

Data Engineer Intern

May 2025 - Aug 2025

Health Alliance Plan

Troy, MI

- Engineered a synthetic data generation pipeline using Python (CTGAN, TVAE, SDV) that produced 10,000+ HIPAA-compliant patient records with 95% similarity to production data.
- Automated medical diagnosis extraction from 5,000+ patient PDF records using Python NLP pipeline (spaCy, PyMuPDF, Tesseract OCR), reducing manual review time from 100+ hours to 8 hours weekly.
- Built interactive analytics dashboards in Jupyter using matplotlib and pandas to visualize prescription trends, patient demographics, and diagnostic patterns across 50K+ patient records, reducing clinician analysis time by 60%.

Software Engineer Intern

May 2024 - Aug 2024

Skyline Web Consultancy

Ann Arbor, MI

- Built 25+ RESTful APIs using Node.js and Express to handle 500K+ monthly requests, reducing average response time from 1.2s to 180 ms through query optimization and Redis caching.
- Accelerated development speed by 40% by building a reusable Node.js microservices framework with standardized error handling, logging, and Swagger API documentation.
- Enhanced system reliability to 99.8% uptime by implementing unit and integration testing, establishing CI/CD pipelines with GitHub Actions, and automating deployments to AWS EC2.
- Integrated third-party payment APIs (Stripe, PayPal) with Datadog monitoring, achieving 99.8% payment-processing uptime and processing \$50K+ in monthly transactions.

Software Engineer Intern

Jan 2024 – Mar 2024

AdduXi

Rochester Hills, MI

- Developed a full-stack production tracking platform using Java (Spring Boot) and React enabling 100+ line workers to log and track production data, reducing manual input errors by 15%.
- Optimized PostgreSQL database architecture and queries, cutting data retrieval times by 65% and enabling real-time analytics across 1M+ production records for supervisors and engineers.

PROJECTS

AI Database Query Platform | Python, FastAPI, React, AWS EC2, PostgreSQL, Docker, Kafka

- Developed an AI-driven NL-to-SQL platform using AWS Bedrock (Claude Sonnet) and FastAPI, enabling non-technical users to query 500K+ equipment records in plain English.
- Built an interactive React dashboard featuring chat-style querying, data visualization, saved searches, and CSV export.

ADDITIONAL

Programming: Python, C++, Java, JavaScript, HTML/CSS, SQL, Node.js, Express, Spring Boot, FastAPI, Flask, React, pandas, NumPy, matplotlib, spaCy, scikit-learn, CTGAN, TVAE, SDV, PyMuPDF, openpyxl, Jinja2

Tools: GGit, GitHub Actions, Jenkins, Docker, Kafka, Redis, MQTT, AWS EC2, AWS Bedrock, Amazon RDS, PostgreSQL, MySQL, Datadog, Swagger, Postman, Power BI, Excel, Qt, STL, Jupyter Notebook, Tesseract OCR, Microsoft Graph API, Nginx, WebSocket